

Lab Assignment 7: Routing Protocols Simulation Report

1. Introduction

The objective of this assignment was to simulate and analyze the fundamental working of four key routing protocols: RIP (Routing Information Protocol), OSPF (Open Shortest Path First), BGP (Border Gateway Protocol), and IS-IS (Intermediate System to Intermediate System).

This report details the implementation of these simulations in Python, using the networkx library for graph representation. The analysis focuses on the core algorithms, the resulting routing tables, and a comparison of each protocol's operational behavior.

2. Network Topology Diagrams

Two distinct network topologies were designed to simulate the different operating environments of these protocols.

2.1 Intra-Domain Topology (for RIP, OSPF, IS-IS)

A single graph was used for Parts 1, 2, and 4 to allow for direct comparison.

- **For RIP (Part 1):** All link costs were treated as 1 (hop count).
- **For OSPF & IS-IS (Parts 2 & 4):** The specified link weights/costs were used.

2.2 Inter-Domain Topology (for BGP)

A separate graph was created to represent Autonomous Systems (ASes) for the Part 3 BGP simulation.

3. Routing Table Snapshots & Analysis

This section presents the final, converged routing tables for each protocol simulation.

3.1 Part 1: RIP (Routing Information Protocol)

Algorithm: RIP, a distance-vector protocol, was simulated using the Bellman-Ford algorithm. The sole metric was **hop count**. Routers were simulated to exchange their full distance vectors until no further changes occurred, achieving convergence.

3.2 Part 2: OSPF (Open Shortest Path First)

Algorithm: OSPF, a link-state protocol, was simulated by assuming the LSA flooding process was complete. Each router, therefore, had a complete map of the network. Each router then

independently ran Dijkstra's algorithm to compute the shortest path tree to all other nodes using the specified **link costs**.

3.3 Part 3: BGP (Border Gateway Protocol)

Algorithm: BGP, a path-vector protocol, was simulated to select routes based on the **shortest AS-path length**. The simulation also implemented BGP's core loop-prevention mechanism by discarding any route update that already contained the router's own AS number.

3.4 Part 4: IS-IS (Intermediate System to Intermediate System)

Algorithm: IS-IS, another link-state protocol, was simulated using the same core logic as OSPF, as specified in the assignment. It builds a complete topology map and uses Dijkstra's algorithm to find the shortest paths based on **link costs**.

4. Comparison of Convergence Behavior and Overhead

The four protocols exhibit significant differences in their convergence speed and the network overhead they generate.

Protocol	Protocol Family	Algorithm	Convergence	Network Overhead
RIP	Distance Vector	Bellman-Ford	Slow. Converges in rounds and is subject to the Count-to-Infinity problem. A change can take many update cycles (iterations) to propagate.	High & Constant. Broadcasts its <i>entire</i> routing table to neighbors periodically (e.g., every 30s), even if no network changes have occurred.
OSPF	Link State	Dijkstra	Fast. Upon a topology change, a lightweight	High during change (LSA flooding). Low when stable,

			Link-State Advertisement (LSA) is flooded. Each router then immediately re-computes its tables locally.	sending only small "hello" packets to maintain neighbor adjacencies.
IS-IS	Link State	Dijkstra	Fast. Functionally identical convergence behavior to OSPF (LSP flooding and local SPF calculation). Often favored by large ISPs for its stability.	High during change (LSP flooding). Low when stable.
BGP	Path Vector	Path Selection (Policy)	Slow. Convergence is also round-based as updates propagate AS-by-AS. Can be subject to "path hunting" or oscillations in complex policies.	Moderate. Only sends <i>updates</i> (new or withdrawn routes), not the full table. Updates are event-triggered. However, a full BGP table itself is very large.

5. Observations and Conclusions

This assignment successfully demonstrated the fundamental differences between the three main families of routing protocols:

1. **Distance-Vector (RIP):** This simulation showed how simple but slow this method is.

Routers rely on "routing by rumor," only trusting what their immediate neighbors tell them. The step-by-step convergence, where the "cost" (hop count) increments one iteration at a time, was clearly visible. Its limitation is that it ignores link-cost and only uses hops.

2. **Link-State (OSPF & IS-IS):** These simulations showed the efficiency of having a complete "map" of the network. By running Dijkstra's algorithm, each router can find the true, cost-based shortest path from its own perspective. This architecture is what makes it converge so much faster and more reliably than RIP. The simulation showed that the algorithmic core of OSPF and IS-IS is identical.
3. **Path-Vector (BGP):** The BGP simulation was fundamentally different. The metric was not cost or hops, but the *path itself* (a list of ASes). This simulation showed the two key BGP principles: path selection (shortest AS path) and loop prevention (discarding paths containing one's own AS). This clearly distinguishes it as a policy-based protocol for inter-domain routing, not just a shortest-path calculator.

In conclusion, the project provided a practical, hands-on understanding of the algorithms that form the backbone of network routing, from simple, small-network protocols like RIP to the complex, policy-driven engine of BGP.